KODAIRA DIMENSION OF FIBRATIONS AND DETERMINANTS OF PLURICANONICAL DIRECT IMAGES

JIHAO LIU

Abstract. We prove a conjecture of Meng on fibrations between smooth projective varieties. When the base has non-negative Kodaira dimension, the conjecture expresses the Kodaira dimension of the total space in terms of the Kodaira dimension of a general fiber and the determinants of the pluricanonical direct images. The proof relies on results claimed in recent manuscripts, among them the subadditivity of Kodaira dimensions and the existence of good minimal models.

Contents

1. Introduction 1
2. Preliminaries 4
3. Lower bound for the Kodaira dimension 5
4. Nonvanishing of determinants of relative direct images 7
5. Upper bound for the Kodaira dimension 13
6. Proof of the main theorem 16
References 16

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $f : X \to Y$ be a fibration between smooth projective varieties and F a general fiber of f, where fibrations, general fibers, and $\kappa(F)$ are as in Definition 2.1. Iitaka conjectured [Uen75, Conjecture C_n] that

$$\kappa(X) \geq \kappa(F) + \kappa(Y). \tag{1.1}$$

Inequality (1.1) was known in the following cases:

(1) When Y is a curve [Kaw82].
(2) When Y is of general type [Vie82, Kaw81].
(3) When F has a good minimal model [Kaw85].
(4) When F is of general type [Kol87].
(5) When Y is an abelian variety [CP17].
(6) When Y is of maximal Albanese dimension [HPS18].
(7) When Y is a surface [Cao18].
(8) When $\dim X \leq 6$ [Bir09].
(9) When $\dim X \leq 7$, unless $\dim X = 7$, $\dim Y = 3$, and $\kappa(F) = \kappa(Y) = 0$ [Cha25].

Recently, [OAI26a, Theorem 1.1] claimed a proof of Inequality (1.1) for any fibration between smooth projective varieties.

Date: November 10, 2026.
2020 *Mathematics Subject Classification.* 14E30, 14D06, 14C20.
Key words and phrases. Kodaira dimension, Iitaka conjecture, fibration, pluricanonical direct image, determinant line bundle.

Over abelian varieties, Meng proved an equality. Let $f : X \to A$ be a fibration from a smooth projective variety X to an abelian variety A and F a general fiber of f. Then by [Men24, Corollary 1.2],

$$\kappa(X) = \kappa(F) + \kappa\big(A, \widehat{\det}(f_*\omega_X^{\otimes m})\big)$$

for any integer $m \geq 2$ such that $f_*\omega_X^{\otimes m} \neq 0$, where $\widehat{\det}$ is the reflexive determinant (Definition 2.2). Since $\omega_A \cong \mathcal{O}_A$, replacing $\widehat{\det}(f_*\omega_X^{\otimes m})$ by $\widehat{\det}(f_*\omega_X^{\otimes m}) \otimes \omega_A$ does not change the last term, and Meng conjectured the corresponding equality over any base Y with $\kappa(Y) \geq 0$.

Conjecture 1.1 ([Men24, Conjecture 1.8])**.** *Let $f : X \to Y$ be a fibration between smooth projective varieties and F a general fiber of f. Assume that $\kappa(Y) \geq 0$. Then there exists a positive integer N such that*

$$\kappa(X) = \kappa(F) + \kappa\big(Y, \widehat{\det}(f_*\omega_X^{\otimes m}) \otimes \omega_Y\big)$$

for any positive integer m divisible by N such that $f_*\omega_X^{\otimes m} \neq 0$.

Conjecture 1.1 was known in the following cases:

(1) When Y is of maximal Albanese dimension [Men24].
(2) When Y is of general type [Men24].

The goal of this paper is to prove Conjecture 1.1 in full generality.

Theorem 1.2. *Conjecture 1.1 holds.*

Remark 1.3. The proof of Theorem 1.2 depends on the following results, which are claimed in AI-generated manuscripts made public on October 6, 2026 at `https://github.com/openai/math` and have not been refereed.

(1) The inequality (1.1) [OAI26a, Theorem 1.1], used in Sections 3 and 6.
(2) The equality $\kappa(X) = \kappa(F) + \dim Y$ for any fibration $f : X \to Y$ between smooth projective varieties with general fiber F, provided that $NK_X - f^*H$ is pseudo-effective for some positive integer N and some ample divisor H on Y [OAI26c, Corollary 1.3], used in Section 3. The equality $\kappa(X) = \kappa(F) + \dim Y$ is proved in [Sch25, Theorem 7] when $\kappa(F) \geq 0$ and K_Y is pseudo-effective, and in [Zou25, Theorems 1.3 and 5.1] when $\kappa(F) = 0$.
(3) The existence of a good log minimal model of any projective lc pair (X, B) such that $K_X + B$ is pseudo-effective [OAI26b, Theorem 11.1], used in Section 5, and the case $B = 0$ with X smooth [OAI26b, Corollary 11.2], used in Section 4.

Sketch of the proof. Let $f : X \to Y$ and F be as in Conjecture 1.1, so that $\kappa(Y) \geq 0$. For any positive integer m such that $f_*\omega_X^{\otimes m} \neq 0$, let $r_m := \operatorname{rank} f_*\omega_X^{\otimes m}$ and let L_m be a Cartier divisor on Y such that $\mathcal{O}_Y(L_m) \cong \widehat{\det}(f_*\omega_X^{\otimes m})$, as in Set-up 2.3. Then

$$\mathcal{O}_Y(K_Y + L_m) \cong \widehat{\det}(f_*\omega_X^{\otimes m}) \otimes \omega_Y,$$

so the equality in Conjecture 1.1 reads

$$\kappa(X) = \kappa(F) + \kappa(Y, K_Y + L_m).$$

We may assume that $\kappa(F) \geq 0$, since otherwise $f_*\omega_X^{\otimes m} = 0$ for any positive integer m. Then $\kappa(X) \geq \kappa(F) + \kappa(Y) \geq 0$, as Theorem 5.1 requires, by Theorem 2.4, which is Remark 1.3(1). The proof combines the following three results.

(1) (Theorem 3.1) For any positive integer m such that $f_*\omega_X^{\otimes m} \neq 0$,

$$\kappa(X) \geq \kappa(F) + \kappa(Y, K_Y + L_m).$$

(2) (Theorem 4.1) For any integer $m \geq 2$ such that $f_*\omega_{X/Y}^{\otimes m} \neq 0$,

$$\kappa\big(Y, \widehat{\det}(f_*\omega_{X/Y}^{\otimes m})\big) \geq 0.$$

(3) (Theorem 5.1) There exists a positive integer N_0 such that

$$\kappa(X) \le \kappa(F) + \kappa(Y, K_Y + L_m)$$

for any positive integer m divisible by N_0 such that $f_*\omega_X^{\otimes m} \neq 0$ and $\kappa(Y, L_m) \ge 0$.

Let m be a positive integer divisible by $2N_0$ such that $f_*\omega_X^{\otimes m} \neq 0$. Then $m \ge 2$, and by the projection formula,

$$\mathcal{O}_Y(L_m - mr_m K_Y) \cong \widehat{\det}(f_*\omega_{X/Y}^{\otimes m}),$$

so Theorem 4.1 and $\kappa(Y) \ge 0$ imply that $\kappa(Y, L_m) \ge 0$. Thus Theorems 3.1 and 5.1 imply the equality (Section 6).

For Theorem 3.1, Lemma 3.2, proved by [CP17, Theorem 3.4] on a birational modification of f as in [Vie83, Lemma 7.3] and the pseudo-effectivity of K_Y, shows that

$$NK_X - f^*(K_Y + L_m) \text{ is pseudo-effective}$$

for some positive integer N. If $\kappa(Y, K_Y + L_m) > 0$, we follow [Sch25, proof of Proposition 6], with $K_Y + L_m$ in place of K_Y: let $h : Y_1 \to Z$ be the Iitaka fibration of $K_Y + L_m$ on a resolution Y_1 of Y, so that a positive multiple of the pullback of $K_Y + L_m$ is linearly equivalent to $h^*H + A$ for an ample divisor H on Z and a divisor $A \ge 0$, and let $g : X_1 \to Z$ be the composite of h and the induced fibration $X_1 \to Y_1$, where X_1 is a resolution of the component of $X \times_Y Y_1$ dominating Y_1. Then Lemma 3.2 implies the hypothesis of Remark 1.3(2) for g, so

$$\kappa(X) = \kappa(X_z) + \kappa(Y, K_Y + L_m)$$

for $X_z := g^{-1}(z)$ with $z \in Z$ very general. Since $\kappa(Y_z) \ge 0$ for $Y_z := h^{-1}(z)$, Theorem 2.4 for $X_z \to Y_z$ implies $\kappa(X_z) \ge \kappa(F)$.

Previously, $\kappa(Y, L_m) \ge 0$ was known for any positive integer m such that $f_*\omega_X^{\otimes m} \neq 0$ when Y is of maximal Albanese dimension [Men24, Lemma 4.1], and, when the geometric generic fiber of f has a good minimal model,

$$\kappa\big(Y, \widehat{\det}(f_*\omega_{X/Y}^{\otimes m})\big) \ge \mathrm{Var}(f)$$

for some positive integer m [Kaw85, Theorem 1.1(i)], with $\mathrm{Var}(f)$ as in Definition 4.2. Lemma 4.4 passes bigness from one degree to any degree $m \ge 2$.

Lemma 4.3 reduces Theorem 4.1, in each degree $m \ge 2$, to a fibration $h : X' \to Z$ between smooth projective varieties with general fiber F' such that

$$\mathrm{Var}(h) = \dim Z \quad \text{and} \quad \kappa(F') = \kappa(F).$$

The proof of Lemma 4.3 uses a diagram $X \leftarrow \overline{X} \to X'$ over $Y \xleftarrow{\pi} \overline{Y} \xrightarrow{\rho} Z$ as in [Kaw85, p. 6], with π generically finite, and obtains from [Kaw85, Theorem 5.3] in degree m an injection

$$\rho^*\widehat{\det}(h_*\omega_{X'/Z}^{\otimes m}) \hookrightarrow \pi^*\widehat{\det}(f_*\omega_{X/Y}^{\otimes m}) \otimes \mathcal{O}_{\overline{Y}}(R)$$

for a Cartier divisor $R \ge 0$ on $\overline{Y}$ with $\mathrm{codim}_Y \pi(\mathrm{Supp}\, R) \ge 2$, so the norm along the finite part of the Stein factorization of π turns a non-zero section of a power of $\widehat{\det}(h_*\omega_{X'/Z}^{\otimes m})$ into one of a power of $\widehat{\det}(f_*\omega_{X/Y}^{\otimes m})$. Assume that $\dim Z > 0$. Since the bigness of $\widehat{\det}(h_*\omega_{X'/Z}^{\otimes m})$ descends along birational modifications of h [Kaw85, Corollary 5.4], we may assume that h is smooth over $Z \backslash \Sigma$ for a simple normal crossing divisor $\Sigma \subset Z$ such that $h^{-1}(\Sigma)$ is a simple normal crossing divisor. Since $\kappa(F') \ge 0$, Remark 1.3(3) and [Fuj16, Theorem 3.3] imply that the geometric generic fiber of h has a good minimal model, so by [Kaw85, Theorem 1.1(i)], there exists a positive integer n such that

$$\kappa\big(Z, \widehat{\det}(h_*\omega_{X'/Z}^{\otimes n})\big) \ge \mathrm{Var}(h) = \dim Z.$$

Let L and M be Cartier divisors on Z such that

$$\mathcal{O}_Z(L) \cong \widehat{\det}(h_*\omega_{X'/Z}^{\otimes n}) \quad \text{and} \quad \mathcal{O}_Z(M) \cong \widehat{\det}(h_*\omega_{X'/Z}^{\otimes m}),$$

and let $r := \operatorname{rank} h_*\omega_{X'/Z}^{\otimes m}$. Then L is big, and Lemma 4.4 is proved by showing that

$$M - rt\epsilon L \text{ is pseudo-effective}$$

for some rational numbers $t, \epsilon > 0$, so M is big.

As in [CP17, proof of Lemma 5.3], the pseudo-effectivity of $M - rt\epsilon L$ follows from [CP17, Lemma 5.4] with the weight $(m-1-t)\varphi_B + t\varphi_1$ (see Section 4.2) for a sufficiently small rational number $0 < t < m-1$, which exists since $m \geq 2$, where φ_B is the relative m-Bergman weight on $K_{X'/Z}$ [BP08] (see [CP17, Theorem 2.4]) and φ_1 is the weight given by [CP17, Theorem 3.4] in degree n.

For Theorem 5.1, let m be a positive integer divisible by N_0, let $V_m := H^0(X, \omega_X^{\otimes m})$, and let $\mathcal{G}$ be the image of the evaluation map $V_m \otimes \mathcal{O}_Y \to f_*\omega_X^{\otimes m}$. By Lemma 5.2, $\mathcal{G} \neq 0$ and, for a Cartier divisor $L_{\mathcal{G}}$ on Y such that $\mathcal{O}_Y(L_{\mathcal{G}}) \cong \widehat{\det}(\mathcal{G})$,

$$\kappa(Y, L_{\mathcal{G}}) \geq \kappa(X) - \kappa(F).$$

The weak positivity of $f_*\omega_{X/Y}^{\otimes m}$ [Kaw85, Theorem 5.2] and $\kappa(Y) \geq 0$ imply that $K_Y + L_m - L_{\mathcal{G}}$ is pseudo-effective, so by [Nak04, Chapter V, 2.7 Proposition (1) and (2)],

$$\kappa_\sigma(K_Y + L_m) \geq \kappa_\sigma(L_{\mathcal{G}}) \geq \kappa(Y, L_{\mathcal{G}}) \geq \kappa(X) - \kappa(F),$$

where κ_σ is the numerical Iitaka dimension (Definition 5.3). If moreover $\kappa(Y, L_m) \geq 0$, then $\kappa(Y, K_Y + L_m) = \kappa_\sigma(K_Y + L_m)$ by Lemma 5.4, which depends on Remark 1.3(3).

Lemma 5.2 replaces $\dim Y$ in the easy addition $\kappa(X) \leq \kappa(F) + \dim Y$ by $\dim T$, where T is the closure of the image of a dense open subset of Y in a Grassmannian of quotients of V_m, and $\dim T \leq \kappa(Y, L_{\mathcal{G}})$ by the Plücker embedding.

Structure of the paper. In Section 2 we define fibrations and reflexive determinants, and recall the subadditivity of Kodaira dimensions claimed in [OAI26a]. In Section 3 we bound $\kappa(X)$ from below in any degree m such that $f_*\omega_X^{\otimes m} \neq 0$. In Section 4 we prove the non-vanishing of the determinants of the relative pluricanonical direct images in any degree $m \geq 2$. In Section 5 we bound $\kappa(X)$ from above in sufficiently divisible degrees m. In Section 6 we combine the results of Sections 3–5 to complete the proof.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

Definition 2.1. A *fibration* is a projective surjective morphism with connected fibers. Let $f : X \to Y$ be a fibration between varieties. A *general fiber* of f is a fiber $f^{-1}(y)$ with y in a non-empty Zariski open subset of Y depending on the statement at hand, and a *very general fiber* of f is a fiber $f^{-1}(y)$ with y outside a countable union of proper closed subsets of Y depending on the statement at hand. If X and Y are smooth projective varieties and F is a general fiber of f, then we denote by $\kappa(F)$ the Kodaira dimension of a very general fiber of f.

Let $f : X \to Y$ be a fibration between smooth projective varieties and F a general fiber of f. By [Fuj16, Lemma 3.6], any very general fiber F' of f satisfies $\kappa(F') = \kappa(X_{\overline{\eta}})$, where $X_{\overline{\eta}}$ is the geometric generic fiber of f. Thus $\kappa(F) = \kappa(X_{\overline{\eta}})$ does not depend on the choice of the very general fiber, and $X_{\overline{\eta}}$ is the fiber in [OAI26a, Theorem 1.1].

Definition 2.2. Let $\mathcal{E}$ be a torsion-free coherent sheaf of rank $r > 0$ on a smooth variety X. We define

$$\widehat{\det}(\mathcal{E}) := (\wedge^r \mathcal{E})^{**},$$

which is a line bundle on X.

Set-up 2.3. Let X, Y, f, F, r_m, and L_m be as follows.

(1) $f : X \to Y$ is a fibration between smooth projective varieties with general fiber F.
(2) For any positive integer m such that $f_*\omega_X^{\otimes m} \neq 0$, $r_m := \operatorname{rank} f_*\omega_X^{\otimes m}$ and L_m is a Cartier divisor on Y such that $\mathcal{O}_Y(L_m) \cong \widehat{\det}(f_*\omega_X^{\otimes m})$.

Theorem 2.4 ([OAI26a, Theorem 1.1])**.** *Let $f : X \to Y$ be a fibration between smooth projective varieties and F a general fiber of f. Then*

$$\kappa(X) \geq \kappa(F) + \kappa(Y).$$

3. Lower bound for the Kodaira dimension

In this section we prove the following theorem.

Theorem 3.1. *Notation and conditions as in Set-up 2.3. Assume that $\kappa(Y) \geq 0$. Then*

$$\kappa(X) \geq \kappa(F) + \kappa(Y, K_Y + L_m)$$

for any positive integer m such that $f_*\omega_X^{\otimes m} \neq 0$.

Lemma 3.2. *Notation and conditions as in Set-up 2.3. Let m be a positive integer such that $f_*\omega_X^{\otimes m} \neq 0$. Assume that $\kappa(Y) \geq 0$. Then there exists a positive integer N such that $NK_X - f^*(K_Y + L_m)$ is pseudo-effective.*

Proof. **Step 1.** By [Vie83, Lemma 7.3] (see [CP17, Proposition 3.1]), there exist birational morphisms $q : X' \to X$ and $\pi : Y' \to Y$ from smooth projective varieties and a fibration $f' : X' \to Y'$ satisfying the following.

(1) The diagram

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{\pi} & Y \end{array}$$

commutes.
(2) For any prime divisor D on X' such that $\operatorname{codim}_{Y'} f'(D) \geq 2$, we have $\operatorname{codim}_X q(D) \geq 2$.

Possibly replacing Y' and X' with log resolutions, we may assume that f' is smooth over $Y' \backslash \Sigma'$ for a simple normal crossing divisor Σ' on Y' such that $f'^{-1}(\Sigma')$ is a simple normal crossing divisor.

Step 2. Let $U \subset Y$ be the largest open subset over which π is an isomorphism. Then $\operatorname{codim}_Y(Y \backslash U) \geq 2$, so any prime divisor on Y' contained in $Y' \backslash \pi^{-1}(U)$ is π-exceptional. Since $q_*\omega_{X'}^{\otimes m} = \omega_X^{\otimes m}$ and $f \circ q = \pi \circ f'$,

$$f'_*\omega_{X'}^{\otimes m}|_{\pi^{-1}(U)} \cong \pi^*(f_*\omega_X^{\otimes m})|_{\pi^{-1}(U)}.$$

In particular, $\operatorname{rank} f'_*\omega_{X'}^{\otimes m} = r_m$. Let L' be a Cartier divisor on Y' such that

$$\mathcal{O}_{Y'}(L') \cong \widehat{\det}(f'_*\omega_{X'}^{\otimes m}).$$

Then $L'|_{\pi^{-1}(U)} \sim \pi^* L_m|_{\pi^{-1}(U)}$, so possibly replacing L' with a linearly equivalent divisor, we may assume that

$$L' = \pi^* L_m + Q$$

for some divisor Q on Y' whose components are π-exceptional. We let

$$K_{Y'/Y} := K_{Y'} - \pi^* K_Y.$$

Then $K_{Y'/Y} \geq 0$ has positive coefficient along any π-exceptional prime divisor. By the projection formula,

$$f'_*\omega_{X'/Y'}^{\otimes m} \cong f'_*\omega_{X'}^{\otimes m} \otimes \omega_{Y'}^{\otimes -m},$$

which is of rank r_m, so

$$\widehat{\det}(f'_*\omega_{X'/Y'}^{\otimes m}) \cong \mathcal{O}_{Y'}(L' - mr_m K_{Y'}).$$

Step 3. Let $K_{X'/Y'} := K_{X'} - f'^* K_{Y'}$. Since f' is smooth over $Y'\backslash\Sigma'$ and $f'^{-1}(\Sigma')$ is a simple normal crossing divisor, by [CP17, Theorem 3.4] (with trivial twisting line bundle), and since a divisor class on a projective manifold containing a closed positive current is pseudo-effective [Dem12, Theorem 6.17(c)], there exist a real number $\epsilon_0 > 0$ and a $\mathbb{Q}$-divisor $E \geq 0$ on X' such that $\operatorname{codim}_{Y'} f'(\operatorname{Supp} E) \geq 2$ and

$$K_{X'/Y'} + E - \epsilon_0 f'^*(L' - mr_m K_{Y'})$$

is pseudo-effective. By (2), E is q-exceptional. Let F' be a general fiber of f'. Since $f'_*\omega_{X'}^{\otimes m} \neq 0$, $H^0(F', \mathcal{O}_{F'}(mK_{F'})) \neq 0$, so $K_{X'/Y'}$ is pseudo-effective by [BP08] (see [CP17, Theorem 2.4]) and [Dem12, Theorem 6.17(c)]. Since $K_{X'/Y'} + E$ and $K_{X'/Y'} + E - \epsilon_0 f'^*(L' - mr_m K_{Y'})$ are pseudo-effective, so is

$$P := K_{X'/Y'} + E - \frac{1}{N} f'^*(L' - mr_m K_{Y'})$$

for any positive integer N with $\frac{1}{N} \leq \epsilon_0$.

Step 4. Let N be a positive integer such that

$$\frac{1}{N} \leq \epsilon_0, \quad 1 - \frac{mr_m + 1}{N} \geq 0, \quad \text{and} \quad \Big(1 - \frac{mr_m}{N}\Big) K_{Y'/Y} + \frac{1}{N} Q \geq 0.$$

N exists since $K_{Y'/Y}$ has positive coefficient along any component of Q. By the first inequality and **Step 3**, P is pseudo-effective. We let

$$\begin{aligned} D_{Y'} &:= \Big(1 - \frac{mr_m}{N}\Big) K_{Y'} + \frac{1}{N} L' - \frac{1}{N} \pi^*(K_Y + L_m) \\ &= \Big(1 - \frac{mr_m + 1}{N}\Big) \pi^* K_Y + \Big(1 - \frac{mr_m}{N}\Big) K_{Y'/Y} + \frac{1}{N} Q. \end{aligned}$$

Since $\kappa(Y) \geq 0$, $\pi^* K_Y$ is pseudo-effective. By the last two inequalities, $D_{Y'}$ is pseudo-effective, and so is $f'^* D_{Y'}$. Since $K_{X'/Y'} = K_{X'} - f'^* K_{Y'}$ and $\pi \circ f' = f \circ q$,

$$P + f'^* D_{Y'} = K_{X'} + E - \frac{1}{N} q^* f^*(K_Y + L_m).$$

Thus $K_{X'} + E - \frac{1}{N} q^* f^*(K_Y + L_m)$ is pseudo-effective. Since

$$q_* K_{X'} = K_X, \quad q_* E = 0, \quad q_* q^* f^*(K_Y + L_m) = f^*(K_Y + L_m),$$

pushing forward by q, $K_X - \frac{1}{N} f^*(K_Y + L_m)$ is pseudo-effective. The lemma follows. □

Proof of Theorem 3.1. We follow [Sch25, proof of Proposition 6], with $K_Y + L_m$ in place of K_Y. Let m be a positive integer such that $f_*\omega_X^{\otimes m} \neq 0$. Let $d := \kappa(Y, K_Y + L_m)$. If $d = -\infty$, there is nothing to prove. If $d = 0$, by Theorem 2.4,

$$\kappa(X) \geq \kappa(F) + \kappa(Y) \geq \kappa(F).$$

Thus we may assume that $d > 0$.

Step 1. Let $\sigma : Y_1 \to Y$ be a birational morphism and $h : Y_1 \to Z$ a fibration between smooth projective varieties with $\dim Z = d$, giving the Iitaka fibration of $K_Y + L_m$ [Uen75, Theorem 5.10], so that by Kodaira's lemma there exist positive integers c, k, an ample divisor H on Z, and a divisor $A \geq 0$ on Y_1 with

$$kc\sigma^*(K_Y + L_m) \sim h^* H + A.$$

Step 2. Let X_1 be a resolution of the component of $X \times_Y Y_1$ dominating Y_1, with induced morphisms $\mu : X_1 \to X$ and $f_1 : X_1 \to Y_1$, and let $g := h \circ f_1$:

$$\begin{array}{ccc} X_1 & \xrightarrow{\mu} & X \\ {\scriptstyle f_1}\downarrow & & \downarrow{\scriptstyle f} \\ Y_1 & \xrightarrow{\sigma} & Y \\ {\scriptstyle h}\downarrow & & \\ Z & & \end{array} \qquad (g : X_1 \to Z)$$

Then f_1 and g are fibrations, and a very general fiber of f_1 is birational to a very general fiber of f. By Lemma 3.2, there exists a positive integer N such that $NK_X - f^*(K_Y + L_m)$ is pseudo-effective. Since $f \circ \mu = \sigma \circ f_1$ and $K_{X_1} - \mu^* K_X \geq 0$,

$$NK_{X_1} - f_1^*\sigma^*(K_Y + L_m) = \mu^*\big(NK_X - f^*(K_Y + L_m)\big) + N(K_{X_1} - \mu^* K_X)$$

is pseudo-effective. Since $kcf_1^*\sigma^*(K_Y + L_m) \sim g^*H + f_1^*A$ and $f_1^*A \geq 0$, $kcNK_{X_1} - g^*H$ is pseudo-effective. By [OAI26c, Corollary 1.3],

$$\kappa(X) = \kappa(X_1) = \kappa(X_z) + d,$$

where $X_z := g^{-1}(z)$ for a very general point $z \in Z$.

Step 3. Let $Y_z := h^{-1}(z)$. By generic smoothness, Y_z and X_z are smooth projective varieties, and $f_1|_{X_z} : X_z \to Y_z$ is a fibration. Since z is very general, Y_z is not contained in any given proper closed subset of Y_1, so a very general fiber of $f_1|_{X_z}$ is a very general fiber of f_1. Thus the Kodaira dimension of a very general fiber of $f_1|_{X_z}$ is $\kappa(F)$.

Since $\kappa(Y_1) = \kappa(Y) \geq 0$, there exists a positive integer n such that $H^0(Y_1, \mathcal{O}_{Y_1}(nK_{Y_1}))$ contains a non-zero section s. Since z is very general, $s|_{Y_z} \neq 0$. Since $K_{Y_z} = K_{Y_1}|_{Y_z}$, $\kappa(Y_z) \geq 0$. By Theorem 2.4 for $f_1|_{X_z}$,

$$\kappa(X_z) \geq \kappa(F) + \kappa(Y_z) \geq \kappa(F).$$

By **Step 2**,

$$\kappa(X) = \kappa(X_z) + d \geq \kappa(F) + \kappa(Y, K_Y + L_m).$$

The theorem follows. □

4. Nonvanishing of determinants of relative direct images

In this section we prove the following theorem.

Theorem 4.1. *Let $f : X \to Y$ be a fibration between smooth projective varieties. Then*

$$\kappa\big(Y, \widehat{\det}(f_*\omega_{X/Y}^{\otimes m})\big) \geq 0$$

for any integer $m \geq 2$ such that $f_*\omega_{X/Y}^{\otimes m} \neq 0$.

4.1. Reduction to maximal variation.

Definition 4.2 (cf. [Kaw85, Section 1])**.** Let $f : X \to Y$ be a fibration between smooth projective varieties and $\overline{\mathbb{C}(Y)}$ an algebraic closure of the function field of Y. The *variation* $\mathrm{Var}(f)$ of f is the minimum of $\mathrm{trdeg}_{\mathbb{C}} \mathbb{K}$ over all algebraically closed subfields $\mathbb{K} \subset \overline{\mathbb{C}(Y)}$ such that the geometric generic fiber of f is birational to $X_{\mathbb{K}} \times_{\mathbb{K}} \overline{\mathbb{C}(Y)}$ for some variety $X_{\mathbb{K}}$ over $\mathbb{K}$.

Lemma 4.3 (cf. [Kaw85, Section 7, Step 3])**.** *Let $f : X \to Y$ be a fibration between smooth projective varieties and F a general fiber of f. Then there exists a fibration $h : X' \to Z$ between smooth projective varieties with general fiber F' satisfying the following.*

(1) $\mathrm{Var}(h) = \dim Z$ *and* $\kappa(F') = \kappa(F)$.

(2) For any integer $m \geq 2$ such that $f_*\omega_{X/Y}^{\otimes m} \neq 0$, we have $h_*\omega_{X'/Z}^{\otimes m} \neq 0$, and

$$\kappa\big(Z, \widehat{\det}(h_*\omega_{X'/Z}^{\otimes m})\big) \geq 0 \implies \kappa\big(Y, \widehat{\det}(f_*\omega_{X/Y}^{\otimes m})\big) \geq 0.$$

Proof. For a generically finite surjective morphism $g : T \to S$ of varieties, we say, following [Kaw85, p. 4], that a divisor $\Delta \geq 0$ on T is *g-exceptional* if the image under g of any component of Δ has codimension ≥ 2 in S. For a dominant morphism $T \to S$ of varieties and a fibration $X_S \to S$ between normal varieties, the *main component* of $X_S \times_S T$ is the irreducible component dominating T.

Step 1. By Definition 4.2, there exists an algebraically closed subfield $\mathbb{K} \subset \overline{\mathbb{C}(Y)}$ such that $\operatorname{trdeg}_{\mathbb{C}} \mathbb{K} = \operatorname{Var}(f)$ and the geometric generic fiber of f is birational to $X_{\mathbb{K}} \times_{\mathbb{K}} \overline{\mathbb{C}(Y)}$ for some variety $X_{\mathbb{K}}$ over $\mathbb{K}$. By [Kaw85, p. 6], there exist a fibration $h_1 : X_1' \to Z_1$, a generically finite surjective morphism $\pi_1 : \overline{Y}_1 \to Y$, and a surjective morphism $\rho_1 : \overline{Y}_1 \to Z_1$ such that $\overline{\mathbb{C}(Z_1)} = \mathbb{K}$ under the embeddings

$$\mathbb{C}(Z_1) \subset \mathbb{C}(\overline{Y}_1) \subset \overline{\mathbb{C}(Y)}$$

given by ρ_1 and π_1, the main components of $X \times_Y \overline{Y}_1$ and $X_1' \times_{Z_1} \overline{Y}_1$ are birational over $\overline{Y}_1$, and $\operatorname{Var}(h_1) = \operatorname{trdeg}_{\mathbb{C}} \mathbb{K}$. Taking resolutions, we may assume that X_1', Z_1, and $\overline{Y}_1$ are smooth projective. Since $\dim Z_1 = \operatorname{trdeg}_{\mathbb{C}} \mathbb{K}$,

$$\operatorname{Var}(h_1) = \dim Z_1 = \operatorname{Var}(f).$$

By [Vie83, Lemma 7.3], there exist a birational morphism $Z_1^\sharp \to Z_1$ and a resolution $\overline{Y}_1^\sharp$ of the component of $\overline{Y}_1 \times_{Z_1} Z_1^\sharp$ dominating $Z_1^\sharp$, both from smooth projective varieties, such that any prime divisor on $\overline{Y}_1^\sharp$ whose image in $Z_1^\sharp$ has codimension ≥ 2 is contained in the exceptional locus of $\overline{Y}_1^\sharp \to \overline{Y}_1$. The image in Y of such a prime divisor has codimension ≥ 2, since the image in $\overline{Y}_1$ does. Possibly replacing Z_1, $\overline{Y}_1$, and X_1' with $Z_1^\sharp$, $\overline{Y}_1^\sharp$, and a resolution of the main component of $X_1' \times_{Z_1} Z_1^\sharp$, we may assume that

(4.1) any prime divisor P_1 on $\overline{Y}_1$ with $\operatorname{codim}_{Z_1} \rho_1(P_1) \geq 2$ is π_1-exceptional.

Step 2. By [Vie83, Proposition 6.1], there exist an open subset $W \subset Z_1$ such that $\operatorname{codim}_{Z_1}(Z_1 \backslash W) \geq 2$, a finite surjective morphism $W' \to W$ from a smooth variety W', and a resolution X_W' of $X_1' \times_{Z_1} W'$ such that $X_W' \to W'$ is flat with semi-stable fibers. In particular, the fibers of $X_W' \to W'$ are reduced and $X_W' \to W'$ satisfies the normal crossing condition of [Vie83, Lemma 3.6]. Let $\mu : Z \to Z_1$ be the normalization of Z_1 in the function field of W' followed by a resolution which is an isomorphism over W'. Then

$$Z^\circ := \mu^{-1}(W) \cong W'.$$

Let $h : X' \to Z$ be a resolution of a projective compactification of X_W' over Z which is an isomorphism over Z°, so that $h^{-1}(Z^\circ) \cong X_W'$ over Z°. Let $\overline{Y}$ be a resolution of an irreducible component of $\overline{Y}_1 \times_{Z_1} Z$ dominating Z, let $\lambda : \overline{Y} \to \overline{Y}_1$ and $\rho : \overline{Y} \to Z$ be the induced morphisms, and let $\pi := \pi_1 \circ \lambda$. Since μ and λ are generically finite, $\overline{\mathbb{C}(Z)} = \overline{\mathbb{C}(Z_1)}$, the geometric generic fiber of h is the geometric generic fiber of h_1, and the main components of $X \times_Y \overline{Y}$ and $X' \times_Z \overline{Y}$ are birational over $\overline{Y}$. In particular,

$$\operatorname{Var}(h) = \operatorname{Var}(h_1) = \dim Z_1 = \dim Z.$$

Since $\overline{Y}$ is smooth, by [Vie83, Lemma 3.6] with $h^{-1}(Z^\circ) \to Z^\circ$ and $\rho^{-1}(Z^\circ) \to Z^\circ$, there exists an open subset $U \subset Z^\circ$ such that $\operatorname{codim}_{Z^\circ}(Z^\circ \backslash U) \geq 2$ and $h^{-1}(U) \times_U \rho^{-1}(U)$ has only rational singularities. Since $\overline{Y}$ is integral and Z is smooth, ρ is flat over any codimension 1 point of Z. Thus there exists an open subset $Z_0 \subset U$ such that $\operatorname{codim}_{Z^\circ}(Z^\circ \backslash Z_0) \geq 2$ and ρ is flat over Z_0. We show that the following hold.

(a) h and ρ are flat over Z_0, and the fibers of h over Z_0 are reduced.
(b) $X' \times_Z \rho^{-1}(Z_0)$ is a normal variety with only rational singularities.
(c) Any prime divisor P on $\overline{Y}$ such that $\rho(P) \cap Z_0$ is empty or of codimension ≥ 2 in Z_0 is π-exceptional.

(a) holds since $Z_0 \subset Z^\circ$, $h^{-1}(Z^\circ) \to Z^\circ$ is isomorphic to $X'_W \to W'$, which is flat with reduced fibers, and ρ is flat over Z_0 by the choice of Z_0. For (b), $X' \times_Z \rho^{-1}(Z_0) \to \rho^{-1}(Z_0)$ is flat by (a), and the generic fiber of $X' \times_Z \rho^{-1}(Z_0) \to \rho^{-1}(Z_0)$ is integral since the generic fiber of h is geometrically integral. Thus $X' \times_Z \rho^{-1}(Z_0)$ is irreducible. Since $Z_0 \subset U$, $X' \times_Z \rho^{-1}(Z_0)$ has only rational singularities, hence it is normal. For (c), let P be such a prime divisor. Since $\operatorname{codim}_{Z^\circ}(Z^\circ \backslash Z_0) \geq 2$, either $\rho(P) \cap Z^\circ = \emptyset$, or $\dim \rho(P) \leq \dim Z - 2$. In the first case $\mu(\rho(P)) \subset Z_1 \backslash W$, and in the second case $\dim \mu(\rho(P)) \leq \dim Z_1 - 2$. Thus

$$\operatorname{codim}_{Z_1} \rho_1(\lambda(P)) = \operatorname{codim}_{Z_1} \mu(\rho(P)) \geq 2.$$

If $\lambda(P)$ is a prime divisor, then $\operatorname{codim}_Y \pi(P) = \operatorname{codim}_Y \pi_1(\lambda(P)) \geq 2$ by (4.1). Otherwise $\dim \lambda(P) \leq \dim Y - 2$, hence $\dim \pi(P) \leq \dim Y - 2$. This proves (c).

Step 3. Let

$$\overline{Y} \xrightarrow{u} \widetilde{Y} \xrightarrow{v} Y$$

be the Stein factorization of π. Since $\widetilde{Y}$ is normal and u is birational, the closed subset $V_u \subset \widetilde{Y}$ over which u is not an isomorphism has codimension ≥ 2. Let $Y_0 := Y \backslash v(V_u)$. Then $\operatorname{codim}_Y(Y \backslash Y_0) \geq 2$, and $\pi^{-1}(Y_0) \to Y_0$ is a finite surjective morphism between smooth varieties, hence flat. Let $V \subset Y$ be the closure of the set of points of Y_0 over which π is not étale. Possibly replacing X with a log resolution of $(X, f^{-1}(V))$, we may assume that $f^{-1}(V)$ is a simple normal crossing divisor.

Let $\overline{X}$ be a resolution of the closure of the graph of a birational map over $\overline{Y}$ between the main components of $X \times_Y \overline{Y}$ and $X' \times_Z \overline{Y}$, and let $\overline{f} : \overline{X} \to \overline{Y}$ be the induced fibration. We obtain the commutative diagram

$$\begin{array}{ccccc} X & \longleftarrow & \overline{X} & \longrightarrow & X' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \overline{f}} & & \downarrow{\scriptstyle h} \\ Y & \xleftarrow{\pi} & \overline{Y} & \xrightarrow{\rho} & Z. \end{array}$$

Since the geometric generic fibers of f, $\overline{f}$, and h become birational after extending the base fields, and plurigenera of smooth projective varieties are birational invariants and do not change under extension of the base field, the Kodaira dimensions of very general fibers of f, $\overline{f}$, and h are equal [Fuj16, Lemma 3.6], and

$$\operatorname{rank} f_*\omega_{X/Y}^{\otimes m} = \operatorname{rank} \overline{f}_*\omega_{\overline{X}/\overline{Y}}^{\otimes m} = \operatorname{rank} h_*\omega_{X'/Z}^{\otimes m} \quad \text{for any integer } m > 0. \tag{4.2}$$

Thus $\kappa(F') = \kappa(F)$, which proves (1) together with **Step 2**, and $h_*\omega_{X'/Z}^{\otimes m} \neq 0$ for any integer $m \geq 2$ such that $f_*\omega_{X/Y}^{\otimes m} \neq 0$.

Step 4. Fix an integer $m \geq 2$ such that $f_*\omega_{X/Y}^{\otimes m} \neq 0$, and let

$$\mathcal{L} := \widehat{\det}(f_*\omega_{X/Y}^{\otimes m}), \quad \overline{\mathcal{L}} := \widehat{\det}(\overline{f}_*\omega_{\overline{X}/\overline{Y}}^{\otimes m}), \quad \mathcal{L}' := \widehat{\det}(h_*\omega_{X'/Z}^{\otimes m}),$$

which are line bundles by (4.2).

By (a) and (b), $\rho^{-1}(Z_0) \to Z_0$ is a proper flat base change for h over Z_0 such that $X' \times_Z \rho^{-1}(Z_0)$ is normal with only rational singularities, and $\overline{f}^{-1}(\rho^{-1}(Z_0))$ is a resolution of $X' \times_Z \rho^{-1}(Z_0)$. By [Kaw85, Theorem 5.3] (see also [Vie83, Lemma 3.2]) and since the fibers of h over Z_0 are reduced,

$$\overline{f}_*\omega_{\overline{X}/\overline{Y}}^{\otimes m} \cong \rho^* h_*\omega_{X'/Z}^{\otimes m} \quad \text{over } \rho^{-1}(Z_0).$$

Let $Z_0' \subset Z_0$ be the open subset over which $h_*\omega_{X'/Z}^{\otimes m}$ is locally free. Since $h_*\omega_{X'/Z}^{\otimes m}$ is torsion-free and Z is smooth, $\operatorname{codim}_{Z_0}(Z_0 \backslash Z_0') \geq 2$. Over $\rho^{-1}(Z_0')$ both sheaves above are locally free, so $\overline{\mathcal{L}} \cong \rho^*\mathcal{L}'$ over $\rho^{-1}(Z_0')$. Thus there exist divisors $E \geq 0$ and $E' \geq 0$ on $\overline{Y}$ supported in $\overline{Y} \backslash \rho^{-1}(Z_0')$ such that

$$\overline{\mathcal{L}} \otimes \mathcal{O}_{\overline{Y}}(E) \cong \rho^*\mathcal{L}' \otimes \mathcal{O}_{\overline{Y}}(E'). \tag{4.3}$$

For any prime divisor $P \subset \overline{Y}\backslash\rho^{-1}(Z_0')$, we have $\rho(P)\cap Z_0 \subset Z_0\backslash Z_0'$, so P is π-exceptional by (c). Thus E and E' are π-exceptional.

Since $\pi^{-1}(Y_0)\to Y_0$ is flat and the generic fiber of f is geometrically integral, $X\times_Y \pi^{-1}(Y_0)$ is irreducible. Let X_0 be its normalization. Then X_0 is finite over $f^{-1}(Y_0)$ and étale over $f^{-1}(Y_0\backslash V)$, and $f^{-1}(V)$ is a simple normal crossing divisor. By the Abhyankar lemma [GR71, Exposé XIII, Proposition 5.2 and 5.3.0], X_0 has only quotient singularities, hence only rational singularities. Since $\pi^{-1}(Y_0)\to Y_0$ is proper and flat, and $\overline{f}^{-1}(\pi^{-1}(Y_0))$ is a resolution of X_0, [Kaw85, Theorem 5.3] implies that there exists an injection

$$\overline{f}_*\omega_{\overline{X}/\overline{Y}}^{\otimes m} \hookrightarrow \pi^* f_*\omega_{X/Y}^{\otimes m} \quad \text{over } \pi^{-1}(Y_0).$$

Let $Y_0'\subset Y_0$ be the open subset over which $f_*\omega_{X/Y}^{\otimes m}$ is locally free, and let $\overline{Y}'\subset\overline{Y}$ be the open subset over which $\overline{f}_*\omega_{\overline{X}/\overline{Y}}^{\otimes m}$ is locally free. Then $\operatorname{codim}_Y(Y\backslash Y_0')\geq 2$ and $\operatorname{codim}_{\overline{Y}}(\overline{Y}\backslash\overline{Y}')\geq 2$. Over $\pi^{-1}(Y_0')\cap\overline{Y}'$, the injection above is an injection of locally free sheaves, both of rank $\operatorname{rank} f_*\omega_{X/Y}^{\otimes m}$ by (4.2), so the induced homomorphism $\overline{\mathcal{L}}\to\pi^*\mathcal{L}$ is injective there, and it extends to an injection over $\pi^{-1}(Y_0')$. Thus there exist a divisor $E''\geq 0$ supported in $\overline{Y}\backslash\pi^{-1}(Y_0')$ and an injection of line bundles

$$\overline{\mathcal{L}} \hookrightarrow \pi^*\mathcal{L}\otimes\mathcal{O}_{\overline{Y}}(E''). \tag{4.4}$$

Since $\pi(\overline{Y}\backslash\pi^{-1}(Y_0'))\subset Y\backslash Y_0'$, E'' is π-exceptional.

Step 5. Assume that $\kappa(Z,\mathcal{L}')\geq 0$. Then there exist an integer $k>0$ and a non-zero section $\sigma\in H^0(Z,\mathcal{L}'^{\otimes k})$. Since ρ is surjective, $\rho^*\sigma\neq 0$. Multiplying $\rho^*\sigma$ by the canonical section of $\mathcal{O}_{\overline{Y}}(kE')$ and using (4.3), we obtain a non-zero section of $\overline{\mathcal{L}}^{\otimes k}\otimes\mathcal{O}_{\overline{Y}}(kE)$. By (4.4), we obtain a non-zero section of

$$\pi^*\mathcal{L}^{\otimes k}\otimes\mathcal{O}_{\overline{Y}}(R), \quad \text{where } R := k(E+E'').$$

Then R is π-exceptional. Since v is finite, $\dim u(P)=\dim\pi(P)\leq\dim Y-2$ for any component P of R, so R is u-exceptional and $u_*\mathcal{O}_{\overline{Y}}(R)=\mathcal{O}_{\widetilde{Y}}$. By the projection formula,

$$H^0\big(\overline{Y},\pi^*\mathcal{L}^{\otimes k}\otimes\mathcal{O}_{\overline{Y}}(R)\big)=H^0\big(\widetilde{Y},v^*\mathcal{L}^{\otimes k}\big),$$

so $v^*\mathcal{L}^{\otimes k}$ has a non-zero section σ'. Let $r:=\deg v$. The norm of σ' under v is a non-zero rational section of $\mathcal{L}^{\otimes kr}$ whose divisor is the push-forward of the divisor $\operatorname{div}\sigma'\geq 0$ [Ful98, Proposition 1.4(b)]. Hence the norm is a non-zero section of $\mathcal{L}^{\otimes kr}$ on the smooth variety Y. Thus $\kappa(Y,\mathcal{L})\geq 0$. The lemma follows. □

4.2. **Propagation of bigness.**

Lemma 4.4. *Let $f:X\to Y$ be a fibration between smooth projective varieties, let $\Sigma\subset Y$ be a simple normal crossing divisor, and let n be a positive integer. Assume that*

(i) f is smooth over $Y\backslash\Sigma$,
(ii) $f^{-1}(\Sigma)$ is a simple normal crossing divisor, and
(iii) $\widehat{\det}(f_*\omega_{X/Y}^{\otimes n})$ is big.

Then $\widehat{\det}(f_*\omega_{X/Y}^{\otimes m})$ is big for any integer $m\geq 2$ such that $f_*\omega_{X/Y}^{\otimes m}\neq 0$.

Proof. The proof follows [CP17, proof of Lemma 5.3]. Fix an integer $m\geq 2$ such that $f_*\omega_{X/Y}^{\otimes m}\neq 0$, and let $r:=\operatorname{rank} f_*\omega_{X/Y}^{\otimes m}$. Let $K_{X/Y}:=K_X-f^*K_Y$, and let L and M be divisors on Y such that

$$\mathcal{O}_Y(L)\cong\widehat{\det}(f_*\omega_{X/Y}^{\otimes n}) \quad\text{and}\quad \mathcal{O}_Y(M)\cong\widehat{\det}(f_*\omega_{X/Y}^{\otimes m}).$$

We need to show that M is big.

We use singular Hermitian metrics, curvature currents, Lelong numbers, and multiplier ideal sheaves as in [Dem12], and relative Bergman metrics as in [CP17, Section 2]. A singular Hermitian

metric on a line bundle, or on a $\mathbb{Q}$-divisor, is written $e^{-\varphi}$, where φ denotes the local weights: a local section given by a holomorphic function g in a local trivialization has squared norm $|g|^2e^{-\varphi}$. We write $dd^c\varphi$ for the curvature current of $e^{-\varphi}$, $\nu(\varphi,x)$ for the Lelong number of φ at x, and $\mathcal{I}(\varphi)\subset\mathcal{O}_X$ for the multiplier ideal sheaf, whose germs are the holomorphic functions g such that $|g|^2e^{-\varphi}$ is locally integrable.

Let φ_B be the weight on $K_{X/Y}$ such that $e^{-m\varphi_B}$ is the relative m-Bergman metric on $mK_{X/Y}$ [CP17, (2.10)-(2.11)]. By [BP08] (see [CP17, Theorem 2.4]),

$$dd^c\varphi_B\geq 0 \quad \text{on } X.$$

Step 1. By (i), (ii), and [CP17, Theorem 3.4] with trivial twisting line bundle and degree n, there exist a $\mathbb{Q}$-divisor $E\geq 0$ on X such that $\operatorname{codim}_Y f(\operatorname{Supp}E)\geq 2$ and a real number $\epsilon_0>0$ such that the class

$$c_1(K_{X/Y}+E)-\epsilon_0 f^*c_1(L)$$

contains a closed positive current. Let ϵ be a rational number such that $0<\epsilon\leq\epsilon_0$. Since $dd^c\varphi_B+[E]$ is a closed positive current in $c_1(K_{X/Y}+E)$ and

$$c_1(K_{X/Y}+E)-\epsilon f^*c_1(L)=\Big(1-\frac{\epsilon}{\epsilon_0}\Big)c_1(K_{X/Y}+E)+\frac{\epsilon}{\epsilon_0}\big(c_1(K_{X/Y}+E)-\epsilon_0 f^*c_1(L)\big),$$

the $\mathbb{Q}$-divisor $P:=K_{X/Y}+E-\epsilon f^*L$ carries a weight φ_P with $dd^c\varphi_P\geq 0$ [Dem12, Definition 6.2].

Since L is big, there exist an integer $k>0$ and a non-zero section $\sigma\in H^0(Y,\mathcal{O}_Y(kL))$. Then $\varphi_L:=\frac{1}{k}\log|\sigma|^2$ is a weight on L with $dd^c\varphi_L=\frac{1}{k}[\operatorname{div}\sigma]\geq 0$. Let $\varphi_1:=\varphi_P+\epsilon f^*\varphi_L$, which is a weight on $P+\epsilon f^*L=K_{X/Y}+E$. Then

$$dd^c\varphi_1\geq\epsilon f^*dd^c\varphi_L\geq 0. \tag{4.5}$$

In particular, φ_1 is plurisubharmonic in local trivializations. Since $x\to\nu(\varphi_1,x)$ is upper semicontinuous [Dem12, Theorems 2.8(b) and 2.10(b)] and X is compact,

$$c:=\sup_{x\in X}\nu(\varphi_1,x)<+\infty.$$

Step 2. Let t be a rational number such that

$$0<t<m-1 \quad\text{and}\quad tc<1.$$

Let $D:=\lceil tE\rceil$. Then

$$D\geq 0,\quad \operatorname{Supp}D=\operatorname{Supp}E,\quad D-tE\geq 0,$$

and in particular $\operatorname{codim}_Y f(\operatorname{Supp}D)\geq 2$. Let $G:=(m-1)K_{X/Y}+D$. Then

$$G=(m-1-t)K_{X/Y}+t(K_{X/Y}+E)+(D-tE).$$

Let φ_{D-tE} be the weight of the canonical singular metric of the $\mathbb{Q}$-divisor $D-tE\geq 0$, so that $dd^c\varphi_{D-tE}=[D-tE]\geq 0$. Then

$$\varphi:=(m-1-t)\varphi_B+t\varphi_1+\varphi_{D-tE}$$

is a weight on G. Since $m-1-t>0$, $dd^c\varphi_B\geq 0$, and $dd^c\varphi_{D-tE}\geq 0$, (4.5) implies that

$$dd^c\varphi\geq t\epsilon f^*dd^c\varphi_L. \tag{4.6}$$

Step 3. Let $U\subset Y$ be a non-empty open subset such that $U\cap(\Sigma\cup f(\operatorname{Supp}D))=\emptyset$, $f_*\omega_{X/Y}^{\otimes m}$ is locally free over U and commutes with base change to any point $y\in U$. Let $X_y:=f^{-1}(y)$ for $y\in U$. Over U, f is smooth by (i), $D=E=0$, and hence

$$K_{X/Y}+G=mK_{X/Y}\quad\text{and}\quad\varphi=(m-1-t)\varphi_B+t\varphi_1\quad\text{on } f^{-1}(U).$$

Let $W\subset U$ be an open subset and $s\in H^0(W,f_*\omega_{X/Y}^{\otimes m})$. For $y\in W$ and $u\in H^0(X_y,\omega_{X_y}^{\otimes m})$, let

$$\|u\|_y^{2/m}:=\int_{X_y}|u|^{2/m}.$$

By definition [CP17, (2.10)-(2.11)], in a local trivialization of $\omega_{X/Y}^{\otimes m}$ we have

$$e^{m\varphi_B} = \sup\{|u|^2 : u \in H^0(X_y, \omega_{X_y}^{\otimes m}),\ \|u\|_y \le 1\} \quad \text{on } X_y.$$

Thus, on X_y,

$$|s|^2 e^{-m\varphi_B} \le \|s|_{X_y}\|_y^2. \tag{4.7}$$

Since f is smooth over W, the function $y \to \|s|_{X_y}\|_y$ is continuous on W, hence bounded on compact subsets of W. Let $x \in f^{-1}(W)$, and let g be the holomorphic function representing s in a local trivialization of $\omega_{X/Y}^{\otimes m}$ near x. By (4.7), there exist a relatively compact neighborhood Ω of x and a real number $C > 0$ such that g is bounded on Ω and $|g|^2 e^{-m\varphi_B} \le C$ on Ω. Thus $e^{-\varphi_B} \le C^{1/m}|g|^{-2/m}$ outside the zero locus of g, and since $m - 1 - t > 0$,

$$|g|^2 e^{-\varphi} = |g|^2 e^{-(m-1-t)\varphi_B} e^{-t\varphi_1} \le C^{(m-1-t)/m}|g|^{2(1+t)/m} e^{-t\varphi_1}$$

almost everywhere on Ω. Since $tc < 1$, $\nu(\frac{t}{2}\varphi_1, x') < 1$ for any $x' \in X$, so $e^{-t\varphi_1}$ is locally integrable on X [Dem12, Lemma 5.6(a)]. Thus $|g|^2 e^{-\varphi}$ is integrable near x, that is, $g \in \mathcal{I}(\varphi)_x$. Since $\mathcal{I}(\varphi) \subset \mathcal{O}_X$, we obtain

$$f_*\big(\mathcal{O}_X(K_{X/Y} + G) \otimes \mathcal{I}(\varphi)\big) = f_*\mathcal{O}_X(K_{X/Y} + G) \quad \text{over } U. \tag{4.8}$$

Step 4. We have $K_{X/Y} + G = mK_{X/Y} + D$ and

$$f_*\omega_{X/Y}^{\otimes m} \subset f_*\mathcal{O}_X(mK_{X/Y} + D),$$

with equality over $Y \backslash f(\operatorname{Supp} D)$. Thus

$$\operatorname{rank} f_*\mathcal{O}_X(mK_{X/Y} + D) = r,$$

and since $\operatorname{codim}_Y f(\operatorname{Supp} D) \ge 2$ and Y is smooth,

$$\widehat{\det}\big(f_*\mathcal{O}_X(mK_{X/Y} + D)\big) \cong \mathcal{O}_Y(M).$$

By (4.6), (4.8), and [CP17, Lemma 5.4], $\mathcal{O}_Y(M)$ carries a weight φ_M such that

$$dd^c\varphi_M \ge rt\epsilon dd^c\varphi_L.$$

Thus $dd^c(\varphi_M - rt\epsilon\varphi_L)$ is a closed positive current in $c_1(M) - rt\epsilon c_1(L)$, so $M - rt\epsilon L$ is pseudo-effective [Dem12, Theorem 6.17(c)]. Since $rt\epsilon > 0$ and L is big, $M = (M - rt\epsilon L) + rt\epsilon L$ is big. The lemma follows. □

Proof of Theorem 4.1. Fix an integer $m \ge 2$ such that $f_*\omega_{X/Y}^{\otimes m} \ne 0$. Let F be a general fiber of f. Since $\operatorname{rank} f_*\omega_{X/Y}^{\otimes m} = h^0(F, mK_F)$ and $f_*\omega_{X/Y}^{\otimes m} \ne 0$, we have $\kappa(F) \ge 0$. By Lemma 4.3, there exists a fibration $h : X' \to Z$ between smooth projective varieties with general fiber F' such that $\operatorname{Var}(h) = \dim Z$, $\kappa(F') = \kappa(F) \ge 0$, $h_*\omega_{X'/Z}^{\otimes m} \ne 0$, and

$$\kappa\big(Z, \widehat{\det}(h_*\omega_{X'/Z}^{\otimes m})\big) \ge 0 \implies \kappa\big(Y, \widehat{\det}(f_*\omega_{X/Y}^{\otimes m})\big) \ge 0.$$

Thus we only need to show that $\kappa(Z, \mathcal{L}') \ge 0$, where $\mathcal{L}' := \widehat{\det}(h_*\omega_{X'/Z}^{\otimes m})$.

If $\dim Z = 0$, then $\mathcal{L}'$ is the determinant of a non-zero vector space, hence a trivial line bundle.

Assume that $\dim Z > 0$. Let $V \subsetneq Z$ be the closed subset over which h is not smooth, let $\tau : Z'' \to Z$ be a log resolution of (Z, V) with $\Sigma'' := \tau^{-1}(V)$ a simple normal crossing divisor, and let X'' be a log resolution of the main component of $X' \times_Z Z''$ together with the preimage of Σ'', which is an isomorphism over $Z'' \backslash \Sigma''$. Then the induced fibration $h'' : X'' \to Z''$ is smooth over $Z'' \backslash \Sigma''$ and $h''^{-1}(\Sigma'')$ is a simple normal crossing divisor, so h'' satisfies (i) and (ii) of Lemma 4.4 with Σ''. Since τ is birational, $\operatorname{Var}(h'') = \operatorname{Var}(h) = \dim Z''$, very general fibers of h'' are very general fibers of h, whose Kodaira dimension is $\kappa(F) \ge 0$, and

$$\operatorname{rank} h''_*\omega_{X''/Z''}^{\otimes m} = \operatorname{rank} h_*\omega_{X'/Z}^{\otimes m} > 0.$$

By [OAI26b, Corollary 11.2] and [Fuj16, Theorem 3.3], the geometric generic fiber of h'' has a good minimal model, which is a good minimal algebraic variety in the sense of [Kaw85, p. 4]. By [Kaw85, Theorem 1.1(i)], there exists an integer $n > 0$ such that

$$\kappa\big(Z'', \widehat{\det}(h''_*\omega^{\otimes n}_{X''/Z''})\big) \geq \mathrm{Var}(h'') = \dim Z'',$$

that is, $\widehat{\det}(h''_*\omega^{\otimes n}_{X''/Z''})$ is big. By Lemma 4.4, $\widehat{\det}(h''_*\omega^{\otimes m}_{X''/Z''})$ is big. By [Kaw85, Corollary 5.4] with the birational morphism τ, $\mathcal{L}'$ is big. In particular, $\kappa(Z, \mathcal{L}') = \dim Z \geq 0$. □

5. Upper bound for the Kodaira dimension

In this section, we prove the following theorem.

Theorem 5.1. *Notation and conditions as in Set-up 2.3. Assume that $\kappa(X) \geq 0$ and $\kappa(Y) \geq 0$. Then there exists a positive integer N such that*

$$\kappa(X) \leq \kappa(F) + \kappa(Y, K_Y + L_m)$$

for any positive integer m divisible by N such that $f_*\omega^{\otimes m}_X \neq 0$ and $\kappa(Y, L_m) \geq 0$.

Lemma 5.2. *Let $f : X \to Y$ be a fibration between smooth projective varieties and F a general fiber of f. Assume that $\kappa(X) \geq 0$. Then there exists a positive integer N such that for any positive integer m divisible by N, the image $\mathcal{G}$ of the evaluation map*

$$H^0(X, \omega^{\otimes m}_X) \otimes \mathcal{O}_Y \to f_*\omega^{\otimes m}_X$$

is non-zero and

$$\kappa\big(Y, \widehat{\det}(\mathcal{G})\big) \geq \kappa(X) - \kappa(F).$$

Proof. For any $y \in Y$, we let $X_y := f^{-1}(y)$.

Step 1. Since $\kappa(X) \geq 0$, there exists a positive integer n such that $V_n := H^0(X, \omega^{\otimes n}_X) \neq 0$ and the image Z of the rational map $\phi : X \dashrightarrow \mathbb{P}(V_n^*)$ defined by V_n satisfies

$$\dim Z = \kappa(X).$$

Let $\mathrm{Bs}(V_n) \subsetneq X$ be the base locus of V_n. Let $\mu : X' \to X$ be a resolution of the indeterminacy of ϕ which is an isomorphism over $X \backslash \mathrm{Bs}(V_n)$, and $g : X' \to Z$ the induced morphism. We let $W \subset Y \times \mathbb{P}(V_n^*)$ be the image of $(f \circ \mu, g) : X' \to Y \times \mathbb{P}(V_n^*)$ with the reduced structure, and for any $y \in Y$, we let $X'_y := (f \circ \mu)^{-1}(y)$ and W_y be the fiber of $W \to Y$ over y.

The geometric generic fiber of $f \circ \mu$ is integral, and the geometric generic fiber of $W \to Y$ is the image of the geometric generic fiber of $f \circ \mu$, hence is integral. By generic flatness and [Gro66, Théorème 12.2.4 (viii)], there exists a non-empty open subset $U \subset Y$ such that $W_U := W \times_Y U \to U$ is flat and W_y is integral for any $y \in U$. Possibly shrinking U, we may assume that for any $y \in U$, X_y and X'_y are smooth and irreducible, and $X_y \not\subset \mathrm{Bs}(V_n)$. Since $(f \circ \mu, g)(X'_y) = W_y$, X'_y is irreducible, and $\mu^{-1}(X_y \backslash \mathrm{Bs}(V_n))$ is dense in X'_y, we have $W_y = \overline{\phi(X_y \backslash \mathrm{Bs}(V_n))}$ for any $y \in U$. By flatness, $e := \dim W_y$ does not depend on $y \in U$. Let $y \in U$ be a very general point. Since $\omega_X|_{X_y} \cong \omega_{X_y}$, the restriction of ϕ to X_y is the rational map defined by the image of V_n in $H^0(X_y, \omega^{\otimes n}_{X_y})$, so

$$e = \dim W_y \leq \kappa(X_y) = \kappa(F).$$

Step 2. For any integer $q \geq 0$ and any $y \in U$, let

$$I(W_y)_q := \mathrm{Ker}\big(\mathrm{Sym}^q V_n = H^0(\mathbb{P}(V_n^*), \mathcal{O}(q)) \to H^0(W_y, \mathcal{O}_{W_y}(q))\big)$$

be the degree q part of the homogeneous ideal of W_y. In this step, we show that there exists a positive integer N divisible by n such that for any positive integer m divisible by N and any $y \in U$,

$$W_y \text{ is the subscheme of } \mathbb{P}(V_n^*) \text{ cut out by } I(W_y)_{m/n}.$$

Let $\mathcal{I}$ be the ideal sheaf of W_U in $U \times \mathbb{P}(V_n^*)$ and $\pi : U \times \mathbb{P}(V_n^*) \to U$ the projection. Since $\mathcal{O}_{W_U}$ is flat over U, the restriction of $\mathcal{I}$ to $\pi^{-1}(y) = \mathbb{P}(V_n^*)$ is the ideal sheaf $\mathcal{I}_{W_y}$ of W_y for any $y \in U$. By Serre's theorem on each member of a finite affine open cover of U, $\pi^*\pi_*\mathcal{I}(q) \to \mathcal{I}(q)$ is surjective for any $q \gg 0$. Restricting to $\pi^{-1}(y)$, we deduce that there exists a positive integer N divisible by n such that

$$H^0(\mathbb{P}(V_n^*), \mathcal{I}_{W_y}(q)) \otimes \mathcal{O}_{\mathbb{P}(V_n^*)} \to \mathcal{I}_{W_y}(q) \text{ is surjective for any integer } q \geq N/n \text{ and any } y \in U.$$

Since $H^0(\mathbb{P}(V_n^*), \mathcal{I}_{W_y}(q)) = I(W_y)_q$ and $m/n \geq N/n$ for any positive integer m divisible by N, the claim of this step follows.

Step 3. Let m be a positive integer divisible by N, $V_m := H^0(X, \omega_X^{\otimes m})$, and $\mathcal{G}$ the image of $V_m \otimes \mathcal{O}_Y \to f_*\omega_X^{\otimes m}$. For any $0 \neq u \in V_n$, $u^{m/n} \in V_m$ is non-zero, so $\mathcal{G} \neq 0$. Since $\mathcal{G}$ is a subsheaf of the torsion-free sheaf $f_*\omega_X^{\otimes m}$, $\operatorname{rank}\mathcal{G} > 0$ and $\widehat{\det}(\mathcal{G})$ is defined.

There exists a dense open subset $U_m \subset U$ such that $f_*\omega_X^{\otimes m}|_{U_m}$ and $(f_*\omega_X^{\otimes m}/\mathcal{G})|_{U_m}$ are locally free and

$$(f_*\omega_X^{\otimes m}) \otimes k(y) \to H^0(X_y, \omega_X^{\otimes m}|_{X_y})$$

is an isomorphism for any $y \in U_m$. For any $y \in U_m$, let

$$\Lambda_y := \operatorname{Ker}\big(V_m \to H^0(X_y, \omega_X^{\otimes m}|_{X_y})\big).$$

Since $(f_*\omega_X^{\otimes m}/\mathcal{G})|_{U_m}$ is locally free, the homomorphism

$$\mathcal{G} \otimes k(y) \to (f_*\omega_X^{\otimes m}) \otimes k(y)$$

is injective for any $y \in U_m$, hence

$$\Lambda_y = \operatorname{Ker}\big(V_m \to \mathcal{G} \otimes k(y)\big) \quad \text{for any } y \in U_m.$$

Let Gr be the Grassmannian of quotients of V_m of dimension $\operatorname{rank}\mathcal{G}$. Since $\mathcal{G}|_{U_m}$ is a locally free quotient of $V_m \otimes \mathcal{O}_{U_m}$ of rank $\operatorname{rank}\mathcal{G}$, there exists a morphism $\rho : U_m \to \mathrm{Gr}$ which sends any $y \in U_m$ to the quotient $V_m \to \mathcal{G} \otimes k(y)$. Let $T := \overline{\rho(U_m)} \subset \mathrm{Gr}$. Taking the $(\operatorname{rank}\mathcal{G})$-th exterior power and then the double dual of the evaluation map, we obtain a homomorphism

$$\wedge^{\operatorname{rank}\mathcal{G}} V_m \to H^0\big(Y, \widehat{\det}(\mathcal{G})\big),$$

and the rational map defined by the image of $\wedge^{\operatorname{rank}\mathcal{G}} V_m$ is the composition of ρ with the Plücker embedding of Gr over U_m. Thus

$$\kappa\big(Y, \widehat{\det}(\mathcal{G})\big) \geq \dim T.$$

Step 4. Let $c : \operatorname{Sym}^{m/n} V_n \to V_m$ be the multiplication map. We show that

$$c^{-1}(\Lambda_y) = I(W_y)_{m/n} \quad \text{for any } y \in U_m.$$

We regard any $u \in \operatorname{Sym}^{m/n} V_n$ as a homogeneous polynomial of degree m/n on $\mathbb{P}(V_n^*)$. For any $x \in X \backslash \mathrm{Bs}(V_n)$, $c(u)$ vanishes at x if and only if u vanishes at $\phi(x)$. Since X_y is integral and $X_y \backslash \mathrm{Bs}(V_n)$ is dense in X_y, $c(u)|_{X_y} = 0$ if and only if u vanishes on $\phi(X_y \backslash \mathrm{Bs}(V_n))$, if and only if $u \in I(W_y)_{m/n}$.

Let $y, y' \in U_m$ be two points such that $\rho(y) = \rho(y')$. Then $\Lambda_y = \Lambda_{y'}$ by **Step 3**, so $I(W_y)_{m/n} = I(W_{y'})_{m/n}$, and $W_y = W_{y'}$ by **Step 2**. Let $\Gamma \subset \mathrm{Gr} \times \mathbb{P}(V_n^*)$ be the image of $W \times_Y U_m$ under $\rho \times \mathrm{id}_{\mathbb{P}(V_n^*)}$. For any $y \in U_m$, the fiber of Γ over $\rho(y)$ is

$$\bigcup_{y' \in \rho^{-1}(\rho(y))} W_{y'} = W_y,$$

which is of dimension e. Thus $\dim\overline{\Gamma} \leq \dim T + e$. The image of Γ in $\mathbb{P}(V_n^*)$ is $\bigcup_{y \in U_m} W_y \subset Z$, which contains $g((f \circ \mu)^{-1}(U_m))$, and $g((f \circ \mu)^{-1}(U_m))$ is dense in Z. Therefore, by **Steps 1 and 3**,

$$\kappa(X) = \dim Z \leq \dim\overline{\Gamma} \leq e + \dim T \leq \kappa(F) + \kappa\big(Y, \widehat{\det}(\mathcal{G})\big).$$

The lemma follows. □

Definition 5.3 (cf. [Nak04, Chapter V, 2.5 Definition])**.** Let D be a pseudo-effective $\mathbb{R}$-divisor on a smooth projective variety X. The *numerical Iitaka dimension* $\kappa_\sigma(D)$ of D is the largest integer $k \geq 0$ such that

$$\limsup_{m \to +\infty} \frac{h^0(X, \lfloor mD \rfloor + A)}{m^k} > 0$$

for some divisor A on X.

Lemma 5.4. *Let X be a smooth projective variety such that $\kappa(X) \geq 0$, and let L be a Cartier divisor on X such that $\kappa(X, L) \geq 0$. Then $\kappa(X, K_X + L) = \kappa_\sigma(K_X + L)$.*

Proof. Since $\kappa(X) \geq 0$ and $\kappa(X, L) \geq 0$, there exist $\mathbb{Q}$-divisors $D \geq 0$ and $L' \geq 0$ on X such that $D \sim_{\mathbb{Q}} K_X$ and $L' \sim_{\mathbb{Q}} L$. Since X is smooth, there exists a positive integer N such that $(X, \frac{1}{N}L')$ is klt. Since $K_X + \frac{1}{N}L' \sim_{\mathbb{Q}} D + \frac{1}{N}L' \geq 0$, $K_X + \frac{1}{N}L'$ is pseudo-effective. By [OAI26b, Theorem 11.1], $(X, \frac{1}{N}L')$ has a good log minimal model (X', B'), and (X', B') is klt and $X \dashrightarrow X'$ extracts no divisors [OAI26b, Section 2 and Lemma 2.2]. Thus a small $\mathbb{Q}$-factorialization of (X', B') [BCHM10, Corollary 1.4.3] is a good minimal model of $(X, \frac{1}{N}L')$ in the sense of [Fuj16, Definition 3.2], and by [Fuj16, Theorem 3.4],

$$\kappa\big(X, K_X + \tfrac{1}{N}L'\big) = \kappa_\sigma\big(K_X + \tfrac{1}{N}L'\big).$$

Since $D + \frac{1}{N}L' \leq D + L' \leq N(D + \frac{1}{N}L')$, $K_X + L \sim_{\mathbb{Q}} D + L'$, and $K_X + \frac{1}{N}L' \sim_{\mathbb{Q}} D + \frac{1}{N}L'$, and since $\kappa(X, \cdot)$ does not change under multiplication by a positive integer, $\kappa(X, \cdot)$ and κ_σ do not change under $\mathbb{Q}$-linear equivalence and do not decrease when a $\mathbb{Q}$-divisor $E \geq 0$ is added [Nak04, Chapter V, 2.7 Proposition (1)], and

$$\kappa_\sigma\big(N(D + \tfrac{1}{N}L')\big) \leq \kappa_\sigma\big(D + \tfrac{1}{N}L'\big)$$

by restricting the lim sup in Definition 5.3 to the multiples of N, we have

$$\kappa(X, K_X + L) = \kappa\big(X, K_X + \tfrac{1}{N}L'\big) = \kappa_\sigma\big(K_X + \tfrac{1}{N}L'\big) = \kappa_\sigma(K_X + L).$$

The lemma follows. □

Proof of Theorem 5.1. Let N be the positive integer in Lemma 5.2 and m a positive integer divisible by N such that $f_*\omega_X^{\otimes m} \neq 0$ and $\kappa(Y, L_m) \geq 0$. Let $\mathcal{G}$ be the image of the evaluation map

$$H^0(X, \omega_X^{\otimes m}) \otimes \mathcal{O}_Y \to f_*\omega_X^{\otimes m},$$

and $L_{\mathcal{G}}$ a Cartier divisor on Y such that $\mathcal{O}_Y(L_{\mathcal{G}}) \cong \widehat{\det}(\mathcal{G})$. By Lemma 5.2,

$$\kappa(Y, L_{\mathcal{G}}) \geq \kappa(X) - \kappa(F).$$

Let $\mathcal{S}$ be the saturation of $\mathcal{G}$ in $f_*\omega_X^{\otimes m}$ and $\mathcal{Q} := f_*\omega_X^{\otimes m}/\mathcal{S}$, which is torsion-free. Since $\mathcal{G} \subset \mathcal{S}$ and $\operatorname{rank}\mathcal{G} = \operatorname{rank}\mathcal{S}$, there exists a Cartier divisor $D \geq 0$ on Y such that $\widehat{\det}(\mathcal{S}) \cong \mathcal{O}_Y(L_{\mathcal{G}} + D)$. Since Y is smooth and $\mathcal{S}$, $f_*\omega_X^{\otimes m}$, and $\mathcal{Q}$ are locally free outside a closed subset of codimension ≥ 2, we have $L_m \sim L_{\mathcal{G}} + D$ if $\mathcal{Q} = 0$, and $\mathcal{O}_Y(L_m - L_{\mathcal{G}} - D) \cong \widehat{\det}(\mathcal{Q})$ if $\mathcal{Q} \neq 0$.

Assume that $\mathcal{Q} \neq 0$. Since

$$f_*\omega_X^{\otimes m} \otimes \omega_Y^{-m} \cong f_*\omega_{X/Y}^{\otimes m},$$

the sheaf $\mathcal{Q} \otimes \omega_Y^{-m}$ is a non-zero torsion-free quotient of $f_*\omega_{X/Y}^{\otimes m}$. By [Kaw85, Theorem 5.2] (see [Vie83, Theorem III]), $f_*\omega_{X/Y}^{\otimes m}$ is weakly positive [Kaw85, p. 22]. By [Kaw85, Lemma 5.1 and the paragraph after it], the line bundle

$$\widehat{\det}(\mathcal{Q} \otimes \omega_Y^{-m}) \cong \widehat{\det}(\mathcal{Q}) \otimes \omega_Y^{-m \operatorname{rank}\mathcal{Q}}$$

is pseudo-effective, that is,

$$L_m - L_{\mathcal{G}} - D - m(\operatorname{rank}\mathcal{Q})K_Y \text{ is pseudo-effective.}$$

Since $\kappa(Y) \geq 0$ and $D \geq 0$, $L_m - L_{\mathcal{G}}$ is pseudo-effective. If $\mathcal{Q} = 0$, then $L_m - L_{\mathcal{G}} \sim D \geq 0$. Thus in both cases, $L_m - L_{\mathcal{G}}$ is pseudo-effective, and since $\kappa(Y) \geq 0$,

$$(K_Y + L_m) - L_{\mathcal{G}} \text{ is pseudo-effective.}$$

Since $\mathcal{G}$ is generated by global sections, $\kappa(Y, L_{\mathcal{G}}) \geq 0$, so $L_{\mathcal{G}}$ is pseudo-effective. By [Nak04, Chapter V, 2.7 Proposition (1) and (2)] and Lemma 5.2,

$$\kappa_\sigma(K_Y + L_m) \geq \kappa_\sigma(L_{\mathcal{G}}) \geq \kappa(Y, L_{\mathcal{G}}) \geq \kappa(X) - \kappa(F).$$

Since $\kappa(Y) \geq 0$ and $\kappa(Y, L_m) \geq 0$, by Lemma 5.4, $\kappa(Y, K_Y + L_m) = \kappa_\sigma(K_Y + L_m)$. Thus

$$\kappa(X) \leq \kappa(F) + \kappa(Y, K_Y + L_m).$$

The theorem follows. □

6. Proof of the main theorem

Proof of Theorem 1.2. Let $f : X \to Y$, F, r_m, and L_m be as in Set-up 2.3, and assume that $\kappa(Y) \geq 0$.

First assume that $\kappa(F) = -\infty$. Let m be a positive integer. Then $h^0(F, mK_F) = 0$. Since $f_*\omega_X^{\otimes m}$ is torsion-free of rank $h^0(F, mK_F)$, we have

$$f_*\omega_X^{\otimes m} = 0.$$

Thus there exists no positive integer m such that $f_*\omega_X^{\otimes m} \neq 0$, and Conjecture 1.1 holds for f with $N = 1$.

Now assume that $\kappa(F) \geq 0$. By Theorem 2.4,

$$\kappa(X) \geq \kappa(F) + \kappa(Y) \geq 0.$$

Since $\kappa(X) \geq 0$ and $\kappa(Y) \geq 0$, let N_0 be the positive integer in Theorem 5.1. Let $N := 2N_0$, and let m be a positive integer divisible by N such that $f_*\omega_X^{\otimes m} \neq 0$. Then $m \geq 2$ and $N_0 \mid m$.

By the projection formula,

$$f_*\omega_X^{\otimes m} \cong f_*\omega_{X/Y}^{\otimes m} \otimes \omega_Y^{\otimes m}.$$

Thus $f_*\omega_{X/Y}^{\otimes m} \neq 0$, $\operatorname{rank} f_*\omega_{X/Y}^{\otimes m} = r_m$, and

$$\mathcal{O}_Y(L_m - mr_mK_Y) \cong \widehat{\det}(f_*\omega_{X/Y}^{\otimes m}).$$

Since $m \geq 2$, Theorem 4.1 implies that $\kappa(Y, L_m - mr_mK_Y) \geq 0$. Since $\kappa(Y) \geq 0$, we have $\kappa(Y, L_m) \geq 0$. By Theorem 5.1 and Theorem 3.1,

$$\kappa(X) = \kappa(F) + \kappa(Y, K_Y + L_m).$$

Since

$$\mathcal{O}_Y(K_Y + L_m) \cong \widehat{\det}(f_*\omega_X^{\otimes m}) \otimes \omega_Y,$$

the theorem follows. □

References

[BP08] B. Berndtsson and M. Păun, *Bergman kernels and the pseudoeffectivity of relative canonical bundles*, Duke Math. J. **145** (2008), no. 2, 341–378.

[Bir09] C. Birkar, *The Iitaka conjecture $C_{n,m}$ in dimension six*, Compos. Math. **145** (2009), no. 6, 1442–1446.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[Cao18] J. Cao, *Kodaira dimension of algebraic fiber spaces over surfaces*, Algebr. Geom. **5** (2018), no. 6, 728–741.

[CP17] J. Cao and M. Păun, *Kodaira dimension of algebraic fiber spaces over abelian varieties*, Invent. Math. **207** (2017), no. 1, 345–387.

[Cha25] C.-K. Chang, *Generalized nonvanishing conjecture and Iitaka conjecture*, Math. Z. **310** (2025), Paper No. 30.

[Dem12] J.-P. Demailly, *Analytic methods in algebraic geometry*, Surveys of Modern Mathematics 1, International Press, Somerville, MA; Higher Education Press, Beijing (2012).

[Fuj16] O. Fujino, *Direct images of relative pluricanonical bundles*, Algebr. Geom. **3** (2016), no. 1, 50–62.
[Ful98] W. Fulton, *Intersection theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge 2, Springer-Verlag, Berlin (1998).
[Gro66] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. III*, Inst. Hautes Études Sci. Publ. Math. **28** (1966), 5–255.
[GR71] A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Math. **224**, Springer-Verlag, Berlin (1971).
[HPS18] C. D. Hacon, M. Popa, and C. Schnell, *Algebraic fiber spaces over abelian varieties: around a recent theorem by Cao and Păun*, in: *Local and global methods in algebraic geometry*, Contemp. Math. **712**, Amer. Math. Soc., Providence, RI (2018), 143–195.
[Kaw81] Y. Kawamata, *Characterization of abelian varieties*, Compos. Math. **43** (1981), 253–276.
[Kaw82] Y. Kawamata, *Kodaira dimension of algebraic fiber spaces over curves*, Invent. Math. **66** (1982), 57–71.
[Kaw85] Y. Kawamata, *Minimal models and the Kodaira dimension of algebraic fiber spaces*, J. Reine Angew. Math. **363** (1985), 1–46.
[Kol87] J. Kollár, *Subadditivity of the Kodaira dimension: fibers of general type*, in: *Algebraic geometry, Sendai, 1985*, Adv. Stud. Pure Math. **10**, North-Holland, Amsterdam (1987), 361–398.
[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).
[Men24] F. Meng, *On surjective morphisms to abelian varieties and a generalization of the Iitaka conjecture*, Math. Z. **307** (2024), no. 2, Paper No. 28.
[Nak04] N. Nakayama, *Zariski-decomposition and abundance*, MSJ Memoirs **14**, Mathematical Society of Japan, Tokyo (2004).
[OAI26a] OpenAI, *Projective Hodge lines and ordinary Iitaka subadditivity*, Paper No. 033 of https://github.com/openai/math.
[OAI26b] OpenAI, *Log abundance in characteristic zero*, Paper No. 034 of https://github.com/openai/math.
[OAI26c] OpenAI, *Schnell fiber spaces and good canonical models*, Paper No. 034 of https://github.com/openai/math.
[Sch25] C. Schnell, *Singular metrics and a conjecture by Campana and Peternell*, Pure Appl. Math. Q. **21** (2025), no. 3, 1269–1281.
[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.
[Uen75] K. Ueno, *Classification theory of algebraic varieties and compact complex spaces*, Lecture Notes in Math. **439**, Springer-Verlag, Berlin (1975).
[Vie82] E. Viehweg, *Die Additivität der Kodaira Dimension für projektive Faserräume über Varietäten des allgemeinen Typs*, J. Reine Angew. Math. **330** (1982), 132–142.
[Vie83] E. Viehweg, *Weak positivity and the additivity of the Kodaira dimension for certain fibre spaces*, in: *Algebraic varieties and analytic varieties (Tokyo, 1981)*, Adv. Stud. Pure Math. **1**, North-Holland, Amsterdam (1983), 329–353.
[Zou25] Y. Zou, *On the Kodaira dimension of some algebraic fiber spaces*, arXiv:2409.19981.

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Email address: liujihao@math.pku.edu.cn